CS512 Machine Learning
Homework 1: Solutions to Questions 1–4

Hossein Moein

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Report scope: All four questions are answered below. Question 4 numerical results are taken from the executed NumPy implementation on the supplied training and validation splits. The accompanying Jupyter notebook contains the implementation and its outputs.
Note on Report Preparation. I used ChatGPT to assist with LaTeX formatting and organizing my solutions into the required PDF format. I worked through the mathematical derivations and calculations by hand and have retained my handwritten solutions. Images of my handwritten work are also included in the submitted homework materials for additional verification. I can provide the original handwritten pages upon request.

Question 1: AI-assisted Image Annotation (10 points)

Let A, B, C denote the reviewing teams, and let I and R denote an incorrect and correct final label, respectively. Because assignment is equally likely across all 100 annotators, the team priors are proportional to team sizes:

$$P(A) = \frac{50}{100} = 0.50, \quad P(B) = \frac{30}{100} = 0.30, \quad P(C) = \frac{20}{100} = 0.20.$$

The conditional error rates are $P(I \mid A) = 0.04$, $P(I \mid B) = 0.10$, and $P(I \mid C) = 0.25$.

(a) Probability of team A given an incorrect label (5 points)

By the law of total probability,

$$\begin{aligned} P(I) &= P(I \mid A)P(A) + P(I \mid B)P(B) + P(I \mid C)P(C) \\ &= (0.04)(0.50) + (0.10)(0.30) + (0.25)(0.20) \\ &= 0.02 + 0.03 + 0.05 = 0.10. \end{aligned}$$

Applying Bayes' rule,

$$\boxed{P(A \mid I) = \frac{P(I \mid A)P(A)}{P(I)} = \frac{0.04 \cdot 0.50}{0.10} = 0.20 = 20\%.}$$

Thus, among images with incorrect final labels, 20% are expected to have been reviewed by team A.

(b) Probability of team A given a correct label (5 points)

The conditional correct-label rates are

$$P(R \mid A) = 0.96, \qquad P(R \mid B) = 0.90, \qquad P(R \mid C) = 0.75.$$

Hence,

$$\begin{aligned} P(R) &= (0.96)(0.50) + (0.90)(0.30) + (0.75)(0.20) \\ &= 0.48 + 0.27 + 0.15 = 0.90. \end{aligned}$$

By Bayes' rule,

$$\boxed{P(A \mid R) = \frac{P(R \mid A)P(A)}{P(R)} = \frac{0.96 \cdot 0.50}{0.90} = \frac{8}{15} \approx 0.53333 = 53.33\%.}$$

The result differs from part (a) because the three teams have different error rates.

Question 2: Spelling Correction (10 points)

Let W be the intended word and T the observed typed word. We observe $T = \texttt{form}$, and the possible intended words are `from`, `form`, and `foam`.

(a) Bayesian spelling correction (8 points)

The supplied corpus frequencies are $f(\texttt{from}) = 0.004$, $f(\texttt{form}) = 0.0001$, and $f(\texttt{foam}) = 0.00001$. Their sum is

$$Z_f = 0.004 + 0.0001 + 0.00001 = 0.00411.$$

Normalizing over the three candidates gives the prior probabilities

$$P(W = \texttt{from}) = \frac{0.004}{0.00411}, \quad P(W = \texttt{form}) = \frac{0.0001}{0.00411}, \quad P(W = \texttt{foam}) = \frac{0.00001}{0.00411}.$$

The likelihoods of observing the typed word `form` are

$$\begin{aligned} P(T = \texttt{form} \mid W = \texttt{from}) &= 0.02, \\ P(T = \texttt{form} \mid W = \texttt{form}) &= 0.98, \\ P(T = \texttt{form} \mid W = \texttt{foam}) &= 0.004. \end{aligned}$$

Bayes' rule states that

$$P(W = w \mid T = \texttt{form}) = \frac{P(T = \texttt{form} \mid W = w)P(W = w)}{\sum_v P(T = \texttt{form} \mid W = v)P(W = v)}.$$

The common prior normalization factor $1/Z_f$ cancels from numerator and denominator. Thus, the unnormalized posterior weights can be computed directly from the original frequencies:

$$\begin{aligned} q_{\texttt{from}} &= 0.004 \cdot 0.02 = 0.00008000, \\ q_{\texttt{form}} &= 0.0001 \cdot 0.98 = 0.00009800, \\ q_{\texttt{foam}} &= 0.00001 \cdot 0.004 = 0.00000004. \end{aligned}$$

Their sum is $Z_q = 0.00017804$. Therefore,

$$\begin{aligned} \boxed{P(W = \texttt{from} \mid T = \texttt{form})} &= \frac{0.00008000}{0.00017804} \approx 0.449337, \\ \boxed{P(W = \texttt{form} \mid T = \texttt{form})} &= \frac{0.00009800}{0.00017804} \approx 0.550438, \\ \boxed{P(W = \texttt{foam} \mid T = \texttt{form})} &= \frac{0.00000004}{0.00017804} \approx 0.000225. \end{aligned}$$

These probabilities sum to one up to rounding. The largest posterior probability belongs to `form`, so the Bayesian corrector **keeps the typed word** `form`.

(b) Frequency-only spelling correction (2 points)

No: the frequency-only corrector selects `from`, since it has the highest corpus frequency, whereas the Bayesian corrector selects `form`. The Bayesian corrector also incorporates the typing-error probabilities, particularly the 98% chance of typing `form` when `form` was intended, which changes the ranking.

Question 3: Thumbtacks Revisited (16 points)

Let $\theta = P(X = 1)$ denote the probability of heads, with independent Bernoulli observations. For H heads and T tails, $N = H + T$ and the likelihood of the observed sequence is

$$L(\theta) = P(D \mid \theta) = \theta^H (1-\theta)^T, \qquad 0 \le \theta \le 1.$$

(a) Maximum likelihood estimates and normalized likelihood plots (4 points)

For $0 < \theta < 1$, the log likelihood and its derivative are

$$\ell(\theta) = H \log \theta + T \log(1-\theta), \qquad \ell'(\theta) = \frac{H}{\theta} - \frac{T}{1-\theta}.$$

Setting $\ell'(\theta) = 0$ yields $H(1-\theta) = T\theta$, hence

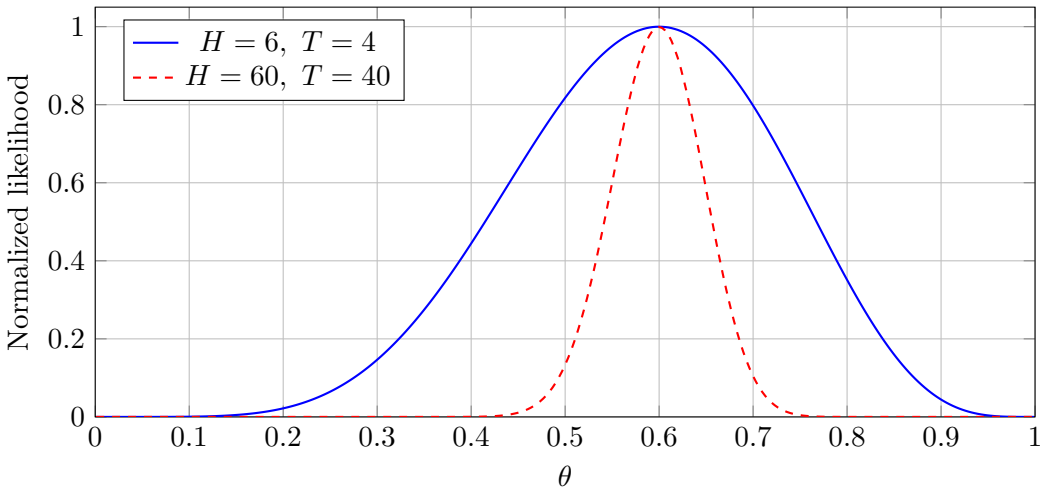
$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{H}{H+T}.}$$

Moreover, $\ell''(\theta) = -H/\theta^2 - T/(1-\theta)^2 < 0$ for these datasets, confirming a unique maximum. Therefore,

$$\boxed{\hat{\theta}_{\text{MLE},1} = \frac{6}{10} = 0.6, \qquad \hat{\theta}_{\text{MLE},2} = \frac{60}{100} = 0.6.}$$

Dividing each likelihood by its own maximum gives

$$\widetilde{L}_1(\theta) = \frac{\theta^6(1-\theta)^4}{(0.6)^6(0.4)^4}, \qquad \widetilde{L}_2(\theta) = \frac{\theta^{60}(1-\theta)^{40}}{(0.6)^{60}(0.4)^{40}}.$$



Interpretation. Both normalized likelihoods peak at $\theta = 0.6$, so their MLEs coincide. However, the 100-toss likelihood is substantially narrower and more concentrated around 0.6 than the 10-toss likelihood. Thus, although the MLE reports the same best-fitting value, the shape of the likelihood reveals that the larger sample provides stronger evidence and greater precision about θ. A normalized likelihood curve is not itself a posterior probability density.

(b) Derivation of the normalized posterior (6 points)

Let α_H and α_T denote the observed head and tail counts, and assume the prior $\theta \sim \text{Beta}(\beta_H, \beta_T)$, with $\beta_H, \beta_T > 0$. The likelihood is

$$P(D \mid \theta) = \theta^{\alpha_H}(1-\theta)^{\alpha_T}.$$

The prior density is

$$p(\theta) = \frac{\theta^{\beta_H - 1}(1-\theta)^{\beta_T - 1}}{B(\beta_H, \beta_T)}, \qquad 0 < \theta < 1.$$

Multiplying likelihood and prior gives

$$P(D \mid \theta)p(\theta) = \frac{1}{B(\beta_H, \beta_T)}\theta^{\alpha_H + \beta_H - 1}(1-\theta)^{\alpha_T + \beta_T - 1}.$$

By Bayes' rule, the normalizing denominator is

$$\begin{aligned} P(D) &= \int_0^1 P(D \mid u)p(u)\, du \\ &= \frac{1}{B(\beta_H, \beta_T)} \int_0^1 u^{\alpha_H + \beta_H - 1}(1-u)^{\alpha_T + \beta_T - 1}\, du \\ &= \frac{B(\alpha_H + \beta_H, \alpha_T + \beta_T)}{B(\beta_H, \beta_T)}, \end{aligned}$$

where the final equality uses $B(a,b) = \int_0^1 u^{a-1}(1-u)^{b-1}\, du$. Dividing by this denominator cancels the original prior normalization constant and yields the *normalized* posterior density:

$$\boxed{p(\theta \mid D) = \frac{\theta^{\alpha_H + \beta_H - 1}(1-\theta)^{\alpha_T + \beta_T - 1}}{B(\alpha_H + \beta_H, \alpha_T + \beta_T)}, \qquad 0 < \theta < 1.}$$

Comparing the exponents and normalization with the Beta density identifies the posterior family and parameters:

$$\boxed{\theta \mid D \sim \text{Beta}(\beta_H + \alpha_H,\ \beta_T + \alpha_T).}$$

Thus, each observed head increments the first Beta shape parameter, and each tail increments the second. The posterior is again Beta, demonstrating that the Beta prior is conjugate to the Bernoulli likelihood.

(c) MAP estimate with a Beta(3,3) prior (6 points)

For $H = 6$ and $T = 4$ with prior Beta$(3,3)$, part (b) gives

$$\theta \mid D \sim \text{Beta}(3+6, 3+4) = \text{Beta}(9,7).$$

Its posterior density is proportional to

$$p(\theta \mid D) \propto \theta^8(1-\theta)^6.$$

Because the normalization constant does not depend on θ, we maximize the log posterior up to an additive constant:

$$g(\theta) = 8\log\theta + 6\log(1-\theta) + C.$$

Differentiating and setting the derivative to zero,

$$g'(\theta) = \frac{8}{\theta} - \frac{6}{1-\theta} = 0 \quad \Longrightarrow \quad 8(1-\theta) = 6\theta \quad \Longrightarrow \quad \boxed{\hat{\theta}_{\text{MAP}} = \frac{8}{14} = \frac{4}{7} \approx 0.57143.}$$

The second derivative, $g''(\theta) = -8/\theta^2 - 6/(1-\theta)^2 < 0$, confirms a maximum. The corresponding MLE is

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{6}{10} = 0.6.}$$

The symmetric Beta$(3, 3)$ prior is centered at 0.5 and favors values near 0.5. Consequently, the MAP estimate 0.57143 is pulled toward 0.5 relative to the data-only MLE 0.6. The posterior mode can also be checked using the Beta mode formula $(a-1)/(a+b-2)$ for $a, b > 1$.

Question 4: Building a Spam Classifier with Naive Bayes (60 points)

Experimental protocol. The dataset contains 700 training emails and 260 validation emails, each represented by a 2500-dimensional bag-of-words count vector. Class 0 denotes non-spam and class 1 denotes spam. All parameters are estimated exclusively from training features and training labels; validation features and labels are used only for evaluation. The multinomial model treats email length as given and assumes conditionally independent word tokens with a class-dependent categorical distribution.

(a) Training-set class distribution (2 points)

There are 350 spam and 350 non-spam emails among 700 training examples. The spam percentage is

$$\boxed{100\,\frac{N_1}{N} = 100\,\frac{350}{700} = 50\%.}$$

Because both classes account for exactly half of the training set, the training-set class distribution is **balanced**; it is not skewed toward either class.

(b) Number of free parameters (5 points)

The two class priors satisfy $\pi_0 + \pi_1 = 1$, so they contribute only $2 - 1 = 1$ independent parameter. For each class $k \in \{0, 1\}$, the $V = 2500$ conditional word probabilities satisfy

$$\sum_{j=1}^{V} \theta_{j|k} = 1,$$

so each class contributes $V - 1 = 2499$ independent parameters. Therefore,

$$\boxed{N_{\text{free}} = (2 - 1) + 2(2500 - 1) = 1 + 4998 = 4999.}$$

The normalization constraints remove one degree of freedom from each of the three probability distributions (one class-prior distribution and two class-conditional word distributions).

(c) MLE classifier and validation evaluation (40 points)

Parameter estimation. Let $t_{j,i}$ be the number of occurrences of vocabulary word j in training email i, and let $y_i \in \{0, 1\}$ be its class. Define the total token count of word j in class k as

$$T_{j,k} = \sum_{i:y_i=k} t_{j,i}, \qquad M_k = \sum_{j=1}^{V} T_{j,k}.$$

The maximum-likelihood class priors and word probabilities are

$$\hat{\pi}_k = \frac{N_k}{N}, \qquad \hat{\theta}_{j|k}^{\text{MLE}} = \frac{T_{j,k}}{M_k}, \quad k \in \{0, 1\}.$$

For this training set, $\hat{\pi}_0 = \hat{\pi}_1 = 350/700 = 0.5$. Importantly, $T_{j,k}$ counts all occurrences, including repeated words within an email, rather than just the number of emails containing the word.

Prediction using log scores. For validation email D_i, define its score for class k by

$$S_k(D_i) = \log \hat{\pi}_k + \sum_{j=1}^{V} t_{j,i} \log \hat{\theta}_{j|k}^{\mathrm{MLE}}.$$

Logarithms convert products of small probabilities into sums and avoid numerical underflow. The conventions required by the assignment are applied explicitly: a term with $t_{j,i} = 0$ contributes zero even when $\hat{\theta}_{j|k} = 0$; if $t_{j,i} > 0$ and $\hat{\theta}_{j|k} = 0$, then $S_k(D_i) = -\infty$. Thus, the implementation first considers only words actually present in the email, and returns $-\infty$ for a class if any present word has zero class-conditional probability. This avoids undefined numerical expressions such as $0 \log 0$.

The prediction rule is

$$\hat{y}_i = \begin{cases} 1, & S_1(D_i) > S_0(D_i), \\ 0, & S_1(D_i) \le S_0(D_i). \end{cases}$$

In particular, equal scores, including two scores of $-\infty$, are resolved in favor of non-spam (class 0).

Validation results and zero-probability diagnosis. The unsmoothed MLE classifier correctly classified 130 of 260 validation emails and misclassified 130:

$$\boxed{\text{Accuracy}_{\mathrm{MLE}} = \frac{130}{260} = 50.00\%, \qquad \text{Incorrect predictions} = 130.}$$

The diagnostic output revealed that *all 260 validation emails* had $S_0(D_i) = S_1(D_i) = -\infty$. Accordingly, the required tie-breaking rule assigned *every* validation email to non-spam (class 0): 260 predicted non-spam and zero predicted spam. Of these, 130 were correct and 130 incorrect. Thus, the 50% accuracy is explained by zero estimated conditional word probabilities and the prescribed tie-breaking behavior, not by successful discrimination between spam and non-spam. The training-set class balance alone does not imply a balanced validation set; here the number of correct class-0 predictions establishes that 130 validation labels were class 0.

(d) MAP estimation with a symmetric Dirichlet prior (8 points)

For each class k, assume the symmetric Dirichlet prior

$$(\theta_{1|k}, \ldots, \theta_{V|k}) \sim \mathrm{Dirichlet}(\alpha + 1, \ldots, \alpha + 1).$$

The likelihood contribution from the class-k training tokens is proportional to $\prod_j \theta_{j|k}^{T_{j,k}}$, while the prior density is proportional to $\prod_j \theta_{j|k}^{\alpha}$. Consequently, the posterior is

$$\boldsymbol{\theta}_k \mid D_{\mathrm{train}} \sim \mathrm{Dirichlet}(T_{1,k} + \alpha + 1, \ldots, T_{V,k} + \alpha + 1).$$

Maximizing this posterior over the probability simplex gives

$$\boxed{\hat{\theta}_{j|k}^{\mathrm{MAP}} = \frac{T_{j,k} + \alpha}{M_k + \alpha V}.}$$

With $\alpha = 1$ and $V = 2500$,

$$\boxed{\hat{\theta}_{j|k}^{\mathrm{MAP}} = \frac{T_{j,k} + 1}{M_k + 2500}, \qquad k \in \{0, 1\}.}$$

Class priors remain the MLEs $\hat{\pi}_0 = \hat{\pi}_1 = 0.5$. Recompute validation log scores and predictions using these smoothed word probabilities, with the same strict comparison $S_1 > S_0$ for predicting spam. Because each smoothed word probability is strictly positive, no present word produces $\log 0$ under this model.

MAP validation results and comparison. With additive smoothing, the classifier correctly predicts 253 of the 260 validation emails and makes seven mistakes:

$$\boxed{\text{Accuracy}_{\text{MAP}} = \frac{253}{260} \times 100\% = 97.31\%, \qquad \text{Incorrect predictions} = 7.}$$

The improvement relative to the unsmoothed MLE classifier is

$$\boxed{97.31\% - 50.00\% = 47.31 \text{ percentage points.}}$$

Smoothing ensures that every class-conditional word probability is positive, so no validation document receives a $-\infty$ score due to an unseen word in either class. Unlike the MLE classifier, which assigned all 260 emails to class 0 because both scores were $-\infty$, the MAP classifier can compare finite class scores and discriminate between the two classes. This explains the substantial observed improvement; it is an empirical comparison on the provided validation split, not a general guarantee that smoothing always increases accuracy.

(e) Five features with largest absolute signed log-ratios (5 points)

Using the smoothed MAP word probabilities from part (d), compute for each vocabulary word

$$\boxed{r_j = \log\left(\frac{\hat{\theta}^{\text{MAP}}_{j|1}}{\hat{\theta}^{\text{MAP}}_{j|0}}\right) = \log \hat{\theta}^{\text{MAP}}_{j|1} - \log \hat{\theta}^{\text{MAP}}_{j|0}.}$$

Rank all 2500 vocabulary indices by $|r_j|$ in descending order and report the five largest magnitudes. The signed value $r_j > 0$ favors spam, $r_j < 0$ favors non-spam, and $r_j = 0$ favors neither. Python arrays are indexed from 0, whereas the assignment requires indices from 1 to 2500; therefore add 1 to the selected array indices before reporting them.

Vocabulary index (1–2500)	Signed log-ratio r_j	Favored class
29	−6.550795	Non-spam
47	−6.270062	Non-spam
88	−5.918518	Non-spam
123	−5.694390	Non-spam
210	−5.437430	Non-spam

Interpretation. A large positive log-ratio indicates a word token much more likely under the spam model, while a large negative value indicates a word token much more likely under the non-spam model. The absolute value measures the strength of this class preference, regardless of direction. All five largest absolute log-ratios are negative; therefore, each of these five features is more probable under the non-spam word distribution than under the spam word distribution. The table reports vocabulary indices rather than word strings because the assignment provides indexed features, not a vocabulary mapping.